Ling 285
Human Language and Technology
Lecture 17

NNs → LLMs → Agentic Systems!!!!

Khalil Iskarous

Department of Linguistics

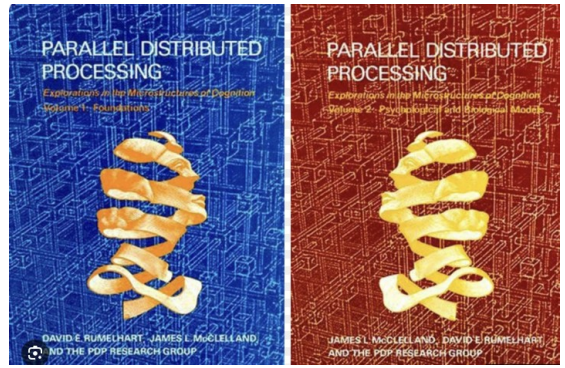
University of Southern California

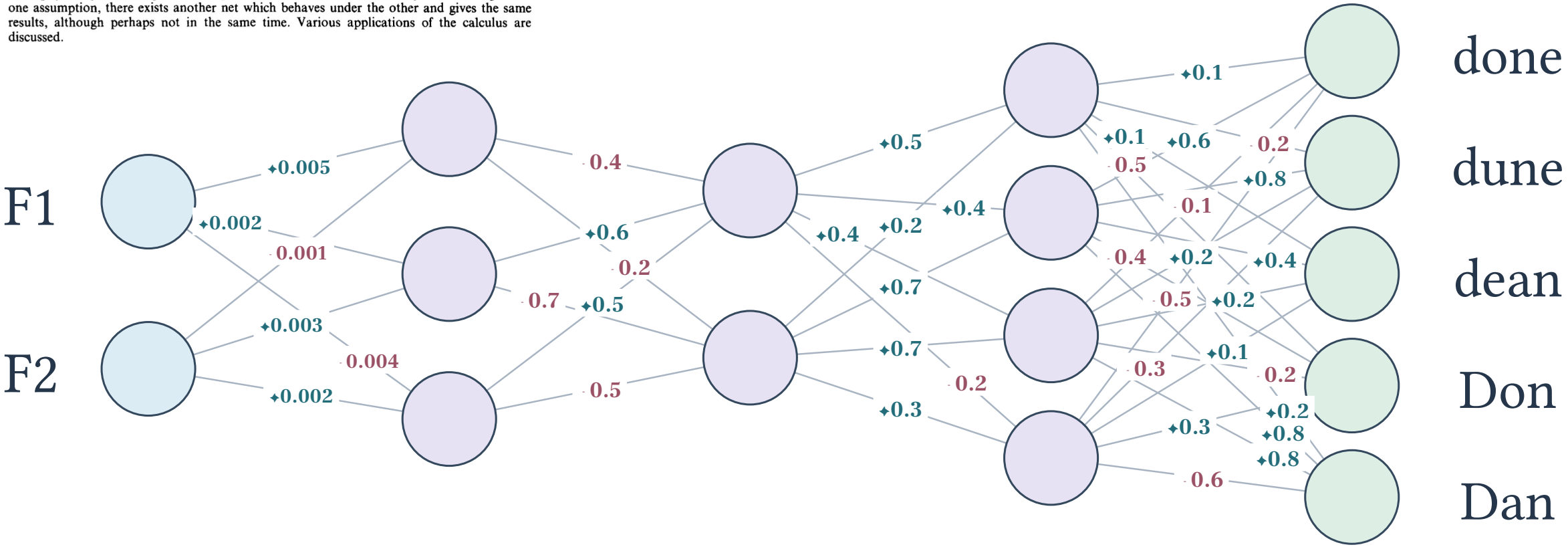
A LOGICAL CALCULUS OF THE IDEAS IMMANENT IN NERVOUS ACTIVITY*

■ WARREN S. MCCULLOCH AND WALTER PITTS
University of Illinois, College of Medicine,
Department of Psychiatry at the Illinois Neuropsychiatric Institute,
University of Chicago, Chicago, U.S.A.

Because of the "all-or-none" character of nervous activity, neural events and the relations among them can be treated by means of propositional logic. It is found that the behavior of every net can be described in these terms, with the addition of more complicated logical means for nets containing circles; and that for any logical expression satisfying certain conditions, one can find a net behaving in the fashion it describes. It is shown that many particular choices among possible neurophysiological assumptions are equivalent, in the sense that for every net behaving under one assumption, there exists another net which behaves under the other and gives the same results, although perhaps not in the same time. Various applications of the calculus are discussed.

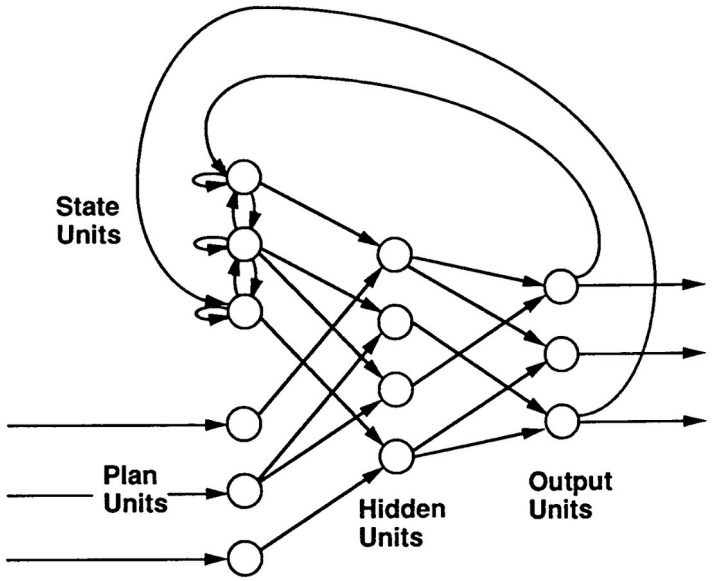
NNs
1943-1986

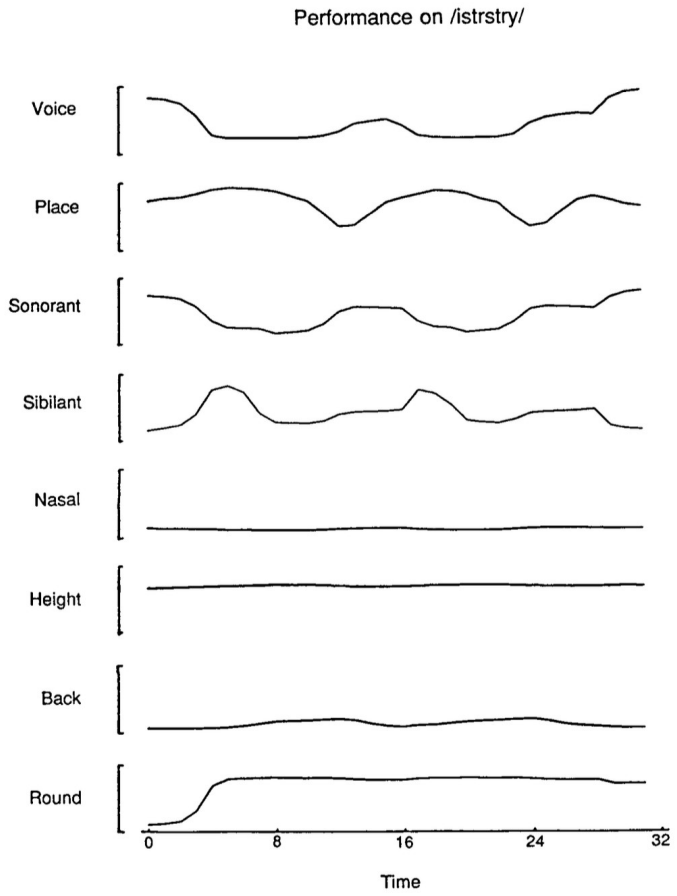




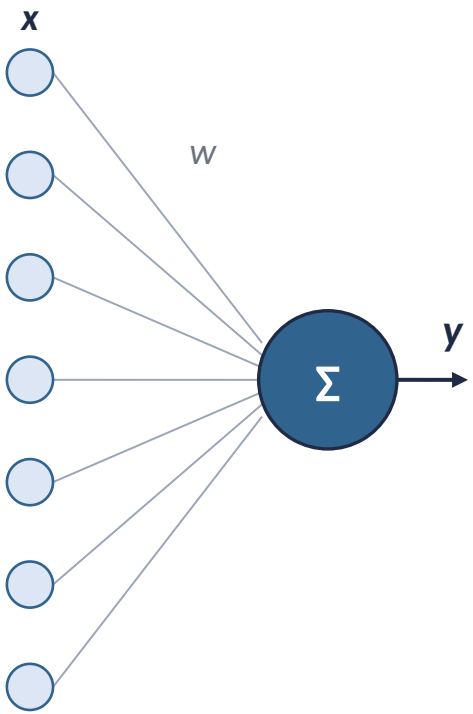
Given some static vector, of formants, the NN could make a decision about a few possible categories… Problem: Human cognition is far more complex…

1985: Michael Jordan developed Recurrent Neural Networks to make sense of speech production…. which became LLMs… then Agentic AI ☺

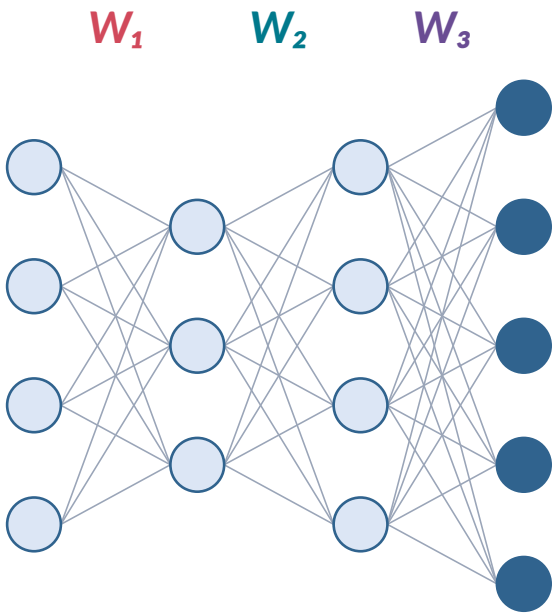








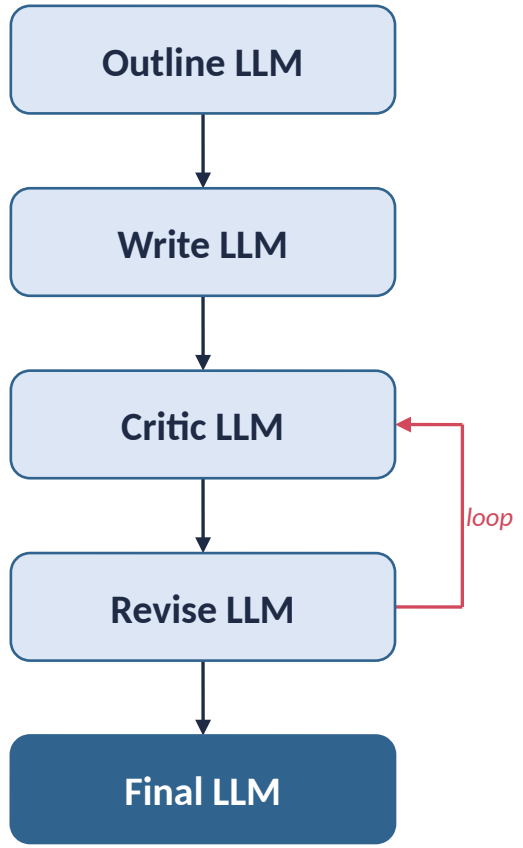
Vector in,
1 number out



Vector in,
Decision vector out



LLM: dynamical system with
many vectors in, **1 decision vector out**

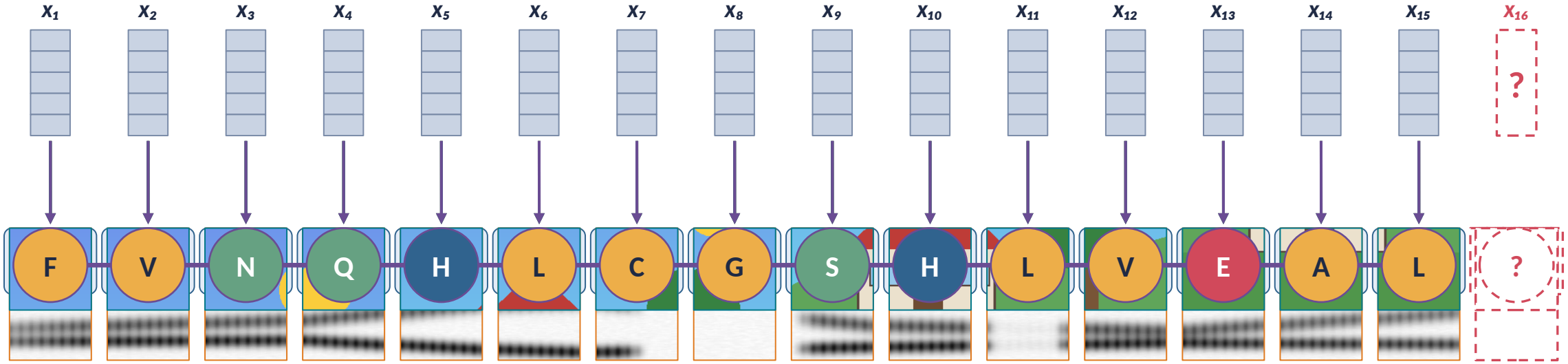


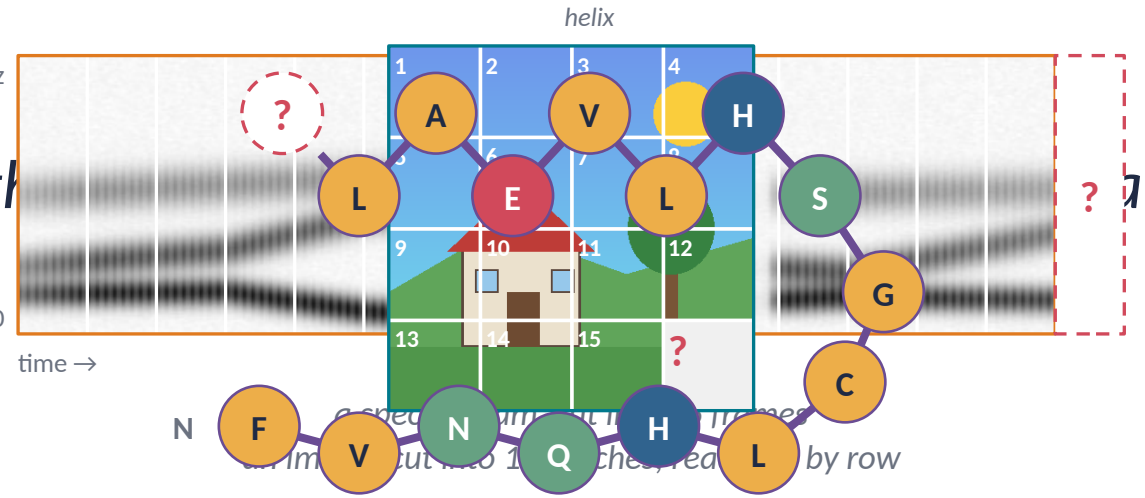
Agentic system:
a graph of LLMs

Inference: inner products

Learning: SGD + DP

15 vectors. *Vectors of what?*



The cat that th  *t on the ____*

a protein (insulin, B chain), read as 15 amino acids

The LLM only ever sees vectors. Words, sounds, pixels, molecules: one machine, one question: what is x_{16}?

Which sequences can be English words?

1 syllable, single C V C

Don	mat	sit	shave	cup	dog	pen	fish	bed	ship
kite	rope	bath	chin	wet	gum	tub	ring	lick	van

A simple syllable in English can have a consonant then a vowel then a consonant

1 syllable, clusters

onsets →					*codas →*				
split	flat	black	glad	stop	desk	box	help	milk	lamp
*sblit	*lfat	*bnack	*dlad	*tsop	*desb	*bokp	*hepl	*mikl	*lapm

Only certain sequences are allowed before the vowel and after the vowel

many syllables

ba·na·na	win·dow	pump·kin	nap·kin	At·lan·tic	ath·lete	ad·mit	spa·ghet·ti
*bnanana			*ngapkin	*tlantic	*thlete	*dmit	*sbaghetti

A long word is a chain of syllables: t·l is fine across a break (At·lan·tic), but tl cannot start a syllable

· syllable break * impossible in English

Pop Quiz!

How phonotactic knowledge can be revealed through GPT learning

Each is the start of an English word. Guess the next letter.

1

ra _

t n r l

after a vowel, almost anything

2

sp _

l r a e i o

l, r or a vowel never ***spn**

3

k _

i e a n

a vowel, or n (knee) never ***kt**

4

skr _

a e i u

only a vowel never ***skrl**

5

m _

a e i o u

only a vowel never ***mb**

6

bram _

b

bramble, or a new syllable

7

b _

r l a e i o u

r, l or a vowel never ***bn**

8

f _

r l a e i o u

r, l or a vowel never ***fn**

9

sh _

a e i o u r

a vowel, or r (shrimp) never ***shb**

Guessing the next letter is predicting x_{16}. A GPT trained only on this game learns English sounds and spelling.

Now we will build an LLM together by designing
an agentic system that learns the
knowledge of English syllables that you have!!!!